

Ocean Odyssey – Assumptions and Design Decisions

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1. Key Conceptual Design Decisions

Address Entity

- **Reason for creation:** A dedicated ADDRESS entity was created instead of embedding address attributes directly in PASSENGER because:
 - Passengers may travel as individuals, families or unrelated groups with distinct addresses.
 - Each passenger has a **unique address instance** and even group travellers may live separately.
- The relationship stayed_in (PASSENGER -> ADDRESS) ensures full traceability of residential information.

Guardian Relationship

- **Unary recursive relationship** (has_guardian) was added within PASSENGER.
 - **Assumption made:** Each minor can have **only one guardian**, but a guardian does not necessarily need to exist for every passenger. The optionality and direction of the relationship support this.
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2. Logical Model Design Choices

Surrogate Keys

Table	Surrogate Key	Reason
PORT_SCHEDULE	port_schedule_id	Replaces composite key (cruise_id, port_datetime) to simplify relationships and improve clarity during joins.
MANIFEST	manifest_id	Added to replace long composite key (cruise_id, passenger_id, ship_code)

ADDRESS	addr_id	Created as a surrogate for complex composite address key (street, town, postcode, country)
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3. Data Types & Validation

- **Standardised VARCHAR/CHAR:**
 - VARCHAR used for names, emails and general string data for more flexibility.
 - CHAR(1) used for pass_gender field.
- **DATE / TIMESTAMP:**
 - Used DATE for cruise/boarding dates and TIMESTAMP for fields like manifest_boarddatetime.

4. Lookup Tables & Country Reuse

Country as Lookup

- COUNTRY is a **lookup table** referenced by:
 - PORT.country_code ~ to track the country each port is located in.
 - SHIP.country_code ~ to register which country each ship is registered under.
 - ADDRESS.country_code ~ allowing foreign key connection from address locations.
- **Why this was done:** Avoids data duplication across SHIP, ADDRESS and PORT to centralise all country names and codes into one consistent table.

5. Relationship Justifications

Relationship	Type	Notes
PASSENGER -> ADDRESS	M:1	Each passenger lives at one address; multiple passengers may share an address.

PASSENGER -> PASSENGER	Unary	Self-FK used to link minors to guardians.
PASSENGER -> MANIFEST	M:N	Passengers can book multiple cruises, stored in MANIFEST.
CABIN -> MANIFEST	1:M	A cabin can be booked by multiple passengers over time.
SHIP -> CABIN	1:M	A ship owns multiple cabins.
CRUISE -> MANIFEST	1:M	Each cruise has many manifest entries.
CRUISE -> PORT_SCHEDULE	1:M	Cruise includes multiple port visits.
PORT_SCHEDULE -> PORT	M:1	Each visit is to one port; ports may be visited multiple times.
